Sisir Kausthubh

Machine Learning | Vision-Language-Action Models (VLAs) | Vision-Language Models (VLMs) | Computer Vision | Reinforcement Learning | Efficient Inference | NLP | Speech Processing

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SUMMARY

I make robot foundation models fast and frugal enough for real robots without losing accuracy, and prove it closed loop. My path runs from perception (computer vision) through decisions (VLMs, agents) to action (robot policies), aimed at frontier robotics: reliable robot brains running on the robot. Every result reports speed, accuracy and efficiency together, against a published baseline.

PROJECTS: VLAS, REINFORCEMENT LEARNING & EMBEDDED

Making an open-source robot model fast (SmolVLA on LIBERO) | efficient-smolvla-voice-to-action Oct 2026 - Present

- **7.7x faster per decision, no retraining:** 111 ms to 14.4 ms median, 147 ms to 16.8 ms p99; **7.6x less energy** (11.0 J to 1.44 J) on an RTX 5090 laptop GPU.
- **Task success held:** 82.00% for both (328 of 400) over the same 400 closed-loop episodes (paired McNemar test, no significant change). Trade-off reported: 152 to 161 s cold startup, 6.5 to 6.7 s warm.
- How: whole-model torch.compile, 1 denoising step instead of 10, batched camera encoding. Tested TensorRT, INT8 and FP8; rejected FP8 vision after success fell to 80%.

Warehouse robot navigation with deep RL BTech

- DDPG continuous-control policy for reaching goals while avoiding obstacles in a simulated Gazebo warehouse; plus TD3 (CNN features) and Q-learning navigation experiments.

Smart home and security system | Raspberry Pi + Arduino BTech

- Live Pi-camera stream (640x480, 24 fps), PIR motion sensor that snaps and emails intruder photos, and a Bluetooth serial door-control prototype on Arduino.

OPEN SOURCE & TALKS

- **Hugging Face LeRobot:** reviewed and benchmarked PR #4634 (SmolVLA KV cache + compile): measured 2.2x end to end on an RTX 5090 laptop; the author added my numbers to the PR. SmolVLA camera-batching change in progress.
- **Talk:** presented this SmolVLA work as a lightning talk, NVIDIA Jetson AI Lab Research Group call, 10 Nov 2026.

PROJECTS: COMPUTER VISION

- **visual-product-tagging:** fine-tuned a 4B vision-language model in 48 minutes on one GPU; 85.3% on ImplicitAVE (GPT-4V: 86.8%) at INR 0.67 per 1,000 products.
- **document-extraction-pipeline:** receipts and GST invoices to checked spreadsheet rows; 84.0 field F1 on CORD (Donut: 83.3 on the same scorer), with business-rule checks and human review.
- **Traffic-sign recognition (GTSRB, 43 classes):** CNN with augmentation compared against a fully connected network, with per-class evaluation.
- **Real-time video style transfer:** feed-forward residual network with VGG16 perceptual loss; foreground and background of a video stylized separately.

PROJECTS: AI AGENTS & GENAI | each runs on one laptop GPU

- **catalog-automation-agent** (LangGraph, MCP server, n8n): auto-tagged 79.2% of 1,610 products at 94.6% accuracy; crash test lost 0 items with 0 duplicates; 45 products/s.
- **docs-rag-assistant:** page-cited RAG with a local 4-bit Gemma 4 26B; 61.8% on FinanceBench (paper's GPT-4-Turbo RAG: 19%).
- **LLM fine-tuning:** Llama 3.2 3B with QLoRA (Unsloth, TRL), trained on assistant replies only, exported to GGUF for local use.
- **AI short-video generator** (Next.js, Remotion): topic to script, generated images, narration and word-timed captions in one workflow.

PROJECTS: LANGUAGE & SPEECH AI | BTech research

- **Tamil-to-Telugu speech-to-speech translation:** Whisper speech recognition (best of 4 models compared on Common Voice 11), IndicTrans2 translation, VITS/MMS speech synthesis.
- **Tamil multimodal hate-speech detection** (final-year research): IndicBERT text and ViT/ResNet video features. **Malayalam headline generation** with a custom encoder-decoder Transformer; Indic document OCR tooling.

PLANNED PROJECTS | designed, starting Nov 2026

GR00T N1.7 fast-path audit (RTX and Jetson) | SmolVLA over ROS 2 | fault-tolerant robot swarms in Isaac Lab | distillation for Jetson

SKILLS

ML techniques: supervised fine-tuning (LoRA/QLoRA), transfer learning, CNNs, Transformers, LSTMs, data augmentation, semi-supervised pseudo-labelling, policy-gradient RL (DDPG, TD3), Q-learning, quantization, time-series forecasting
Inference & ML: Python, PyTorch, torch.compile, TensorRT, INT8/FP8 quantization, vLLM, QLoRA/LoRA fine-tuning, Hugging Face (LeRobot, Transformers, TRL)
VLAs, RL & simulation: VLAs (SmolVLA), LIBERO, MuJoCo, Gazebo, policy-gradient RL (DDPG, TD3), Q-learning, Raspberry Pi, Arduino | learning ROS 2, Isaac Lab, Jetson
Vision, agents & GenAI: vision-language models, CNNs, LangGraph, MCP, RAG, Whisper, paired-statistics evaluation | **Tools:** Linux, Git, uv, Next.js

EDUCATION

BTech, Computer Science and Engineering (AI) | Amrita Vishwa Vidyapeetham Sep 2021 - Jun 2025